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Ab initio study of band offsets in Si/SiO_2 interfaces for electronic applications

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LMAPR2451 : Atomistic and nanoscopic simulations

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Date: November 2, 2026

List of Acronyms

MOSFET Metal–Oxide–Semiconductor Field-Effect Transistor

DFT Density Functional Theory

PECVD Plasma-Enhanced Chemical Vapor Deposition

LPCVD Low Pressure Chemical Vapor Deposition

ALD Atomic Layer Deposition

CPU Central Processing Unit

FFT Fast Fourier Transform

SCF Self-Consistent Field

BFGS Broyden–Fletcher–Goldfarb–Shanno

LDA Local Density Approximation

PBE Perdew-Burke-Ernzerhof

DOS Density of States

EOS Equation Of State

VBO Valence Band Offset

CBO Conduction Band Offset

VBM Valence Band Maximum

CBM Conduction Band Minimum

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1 Introduction

The systematic scaling of electronic technology has placed increasing emphasis on the understanding of material interfaces at the atomic level. Among these interfaces, the silicon/silicon dioxide (Si/SiO_2) interface remains one of the most frequently encountered, as it constitutes one of the main components of modern semiconductor devices such as Metal–Oxide–Semiconductor Field-Effect Transistors (or MOSFETs, whose simplified structure is shown in figure 1). The electronic properties of this interface, especially the band offsets, are crucial in order to determine charge carrier transport, leakage currents and the overall device performance.

In order to study the electronic structure of materials and interfaces, powerful and widely used computational methods such as Density Functional Theory (DFT) have emerged. It enables the calculation of ground-state properties of many-electron systems by solving the Schrödinger equation within the Kohn-Sham framework. In this framework, the electron density is determined by solving the non-interacting Kohn-Sham equations, where an effective potential incorporates the kinetic energy, classical electrostatic interactions, and exchange-correlation effects. The ability of DFT to model realistic interface structures, including defects, strain and atomic rearrangements makes it particularly interesting for analyzing Si/SiO_2 interfaces. Also, by constructing atomistic models of the interface and computing the electronic density of states (DOS), DFT enables to get the band alignment parameters with high accuracy. It is however well known that DFT systematically underestimates band gaps for semiconducting or insulating materials.

This work aims to study the electrostatic potential and subsequently the band offsets at the interface between bulk silicon and silicon dioxide, studied in its β-cristobalite phase, chosen for facility due to its cubic lattice. In typical electronic devices, silicon dioxide is found in thin layers of amorphous phase due to the methods of deposition (e.g. PECVD, LPCVD or ALD), but for the purpose of studying the band offsets at the interface, the choice of β-cristobalite will provide satisfying results.

This work uses the software ABINIT for all calculations. The structure of this report will be the following :

- Convergence studies conducted on separate bulk materials;
- Computed electronic band structures and densities of states of both materials separately;
- Study of the Si/SiO_2 interface including initial supercell construction, structural relaxation and electrical properties;
- Conclusion summarizing the main results.

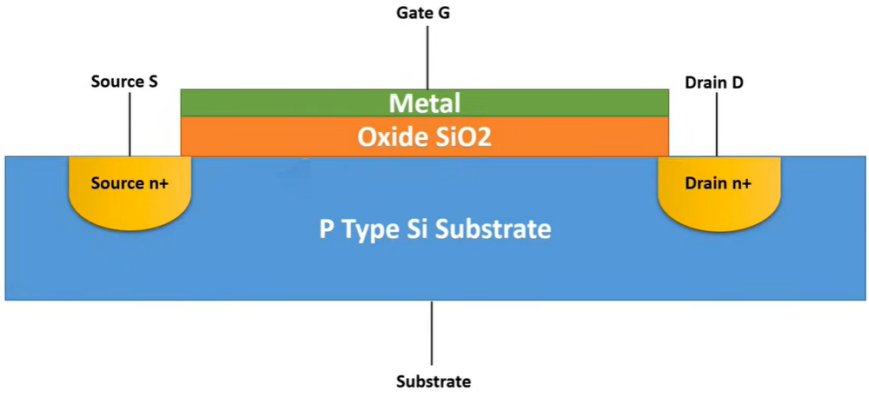


Figure 1: Simplified structure of a NMOS transistor [1].

2 Convergence studies

In this section, we are interested in performing convergence studies on Si and SiO_2 separately to make sure that the chosen numerical settings are accurate for each material. Those convergence studies matter because Si and SiO_2 have different chemical compositions and primitive cells, meaning that they can require different cutoff energies (E_{cut}) and momentum space sampling. Therefore, convergence studies ensure a fair comparison between these crystals. They are also important in order to achieve the best trade-off between computational cost and accuracy. Indeed, convergence studies provide the smallest value of E_{cut} and momentum space sampling for which the calculated results fall within an acceptable threshold, without wasting CPU time.

A convergence study was also conducted for each material with respect to the lattice parameter. In these studies, the lattice parameter (a_{cell}) is sweeped in a plane-wave calculation, which changes the dimensions of the reciprocal lattice and, consequently, the size of the Brillouin Zone. As a result, the number of plane waves can vary during the calculation as the cell expands or contracts. Without smoothing, this would create small, discrete jumps in the total energy. Therefore, trying to find the minimum of the total energy with respect to a_{cell} with these "jiggles" would lead to our optimization getting stuck in a fake numerical well. It is thus important to introduce the cutoff energy smearing (`ecutsm`) parameter.

In plane-wave DFT, increasing E_{cut} means allowing more high-frequency plane waves in the basis set. The number of plane waves N_{pw} is not a continuous function of the cutoff energy, but rather increases in discrete steps as the cutoff radius expands to encompass new shells of reciprocal lattice points. However, the overall curve (if zoomed out to very high energies) roughly follows a $N_{pw} \propto E_{cut}^{3/2}$ relationship. This scaling occurs because the total number of allowed plane waves is proportional to the volume of the cutoff sphere in momentum space. Because this volume grows rapidly as the sphere expands in three dimensions, even a moderate increase in E_{cut} can add many basis functions to the calculation.

Moreover, electrons wavefunctions of our periodic crystals are expanded on those plane waves so that operations on states become heavier as basis size grows. Each Kohn–Sham orbital is expanded as:

$$\psi_{n\mathbf{k}}(\mathbf{r}) = \sum_{\mathbf{G}} c_{n,\mathbf{k}+\mathbf{G}}\, e^{i(\mathbf{k}+\mathbf{G})\cdot\mathbf{r}} \tag{1}$$

where each plane wave is one sinusoidal pattern in space. Small $|\mathbf{G}|$ waves describe smooth, slowly varying parts, whereas large $|\mathbf{G}|$ waves describe rapid variations and fine details. Increasing E_{cut} thus means including more high-$|\mathbf{G}|$ components in equation 1, which is used to represent finer details of the orbitals, density and total energy more accurately. Also, higher E_{cut} requires finer k-point grids to represent density/potential accurately, thus increasing FFT calculation time and memory traffic.

As a result, the total computational cost scales with the number of k-points as well as the number of bands so that the E_{cut} increase is paid repeatedly:

$$\text{Total computational cost} \propto N_k \times N_b \times f(N_{pw}) \tag{2}$$

Therefore, increasing E_{cut} improves basis accuracy but causes an increase in computational cost, runtime and memory.

2.1 Silicon

2.1.1 Cutoff energy

For silicon, figure 2 shows that the changes in the energy per atom become smaller than 0.5 mHa from 12 Ha onwards. Using a larger value of E_{cut} gives only very small improvements, mainly

increasing computational cost. $E_{cut} = 12$ Ha is the value chosen as converged.

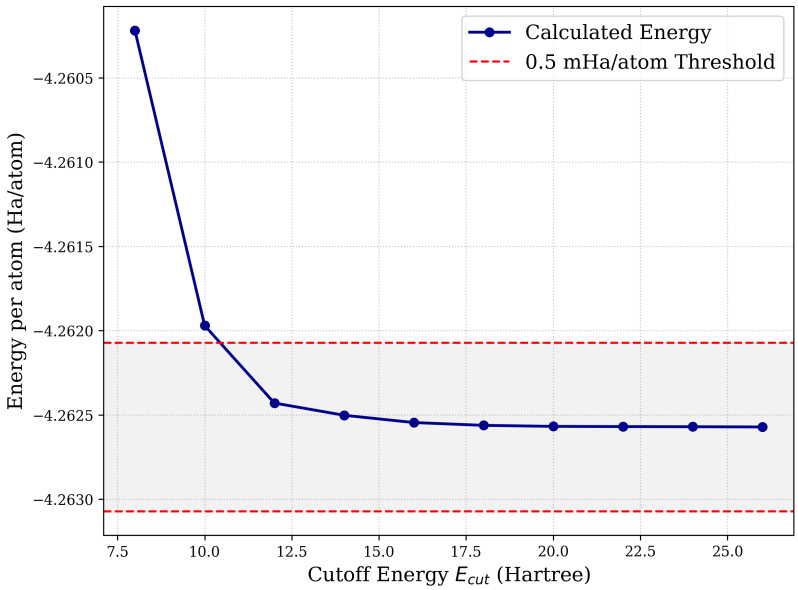


Figure 2: Convergence of the energy per atom as a function of the cut-off energy (E_{cut}) for silicon.

2.1.2 K-point grid

The goal is to have a dense enough sampling in reciprocal space, which is useful to improve Brillouin zone integration accuracy. To do that, it is essential to select a good k-point grid, meaning it must neither be too sparse to have a good sampling nor too dense because computational cost grows with the number of k-points, as explained previously.

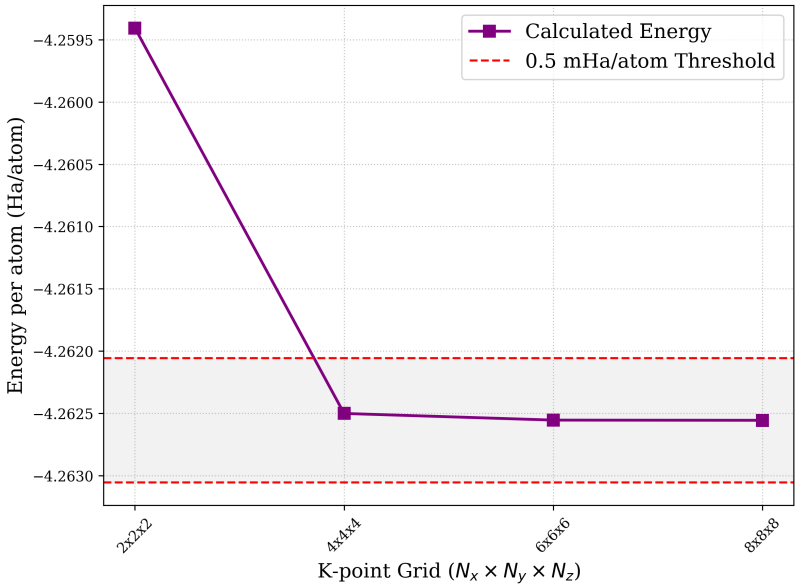


Figure 3: Convergence of the energy per atom as a function of the number of points in the k-point grid for silicon.

For silicon, the grid 4×4×4 is chosen because denser grids only show tiny energy differences with respect to the 4×4×4 grid (see figure 3).

2.1.3 Lattice parameter

Figure 4 shows that the optimized lattice parameter for Si exhibits a characteristic overshoot, peaking at more than 5.396 Å around $E_{cut} = 10$ Ha and then dropping sharply. This behavior is typical for finite basis sets. Indeed, an incomplete basis set can cause artificial lattice expansion initially. For E_{cut} values above 16 Ha, the lattice parameter stabilizes near 5.394 Å with negligible variation (<0.2 mÅ). The red dashed lines mark a 1 mÅ tolerance. From $E_{cut} = 14$ Ha onwards, the parameter stays within this threshold, indicating practical convergence. As a result, we designate the value of the lattice parameter of silicon as 5.394 Å.

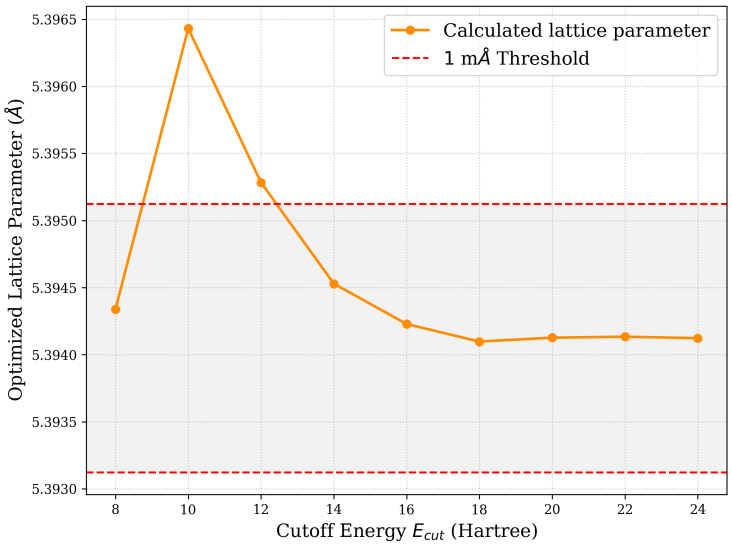


Figure 4: Convergence of the lattice parameter as a function of the cut-off energy (E_{cut}) for silicon, obtained with Broyden–Fletcher–Goldfarb–Shanno (BFGS) optimizer.

This value can be compared with the experimental one (5.431 Å [2]) and the one reported by the Materials Project (5.44 Å [3]). The observed discrepancy with our result can be attributed primarily to the well-known tendency of the Local Density Approximation (LDA) to overbind atoms. Because the LDA functional (Perdew-Wang 1992) used in our work relies solely on the local electron gas density and does not account for density gradients, it systematically predicts stronger, shorter bonds, leading to an underestimation of the lattice parameter by roughly 1%. In contrast, the slightly larger value reported by the Materials Project is characteristic of the Generalized Gradient Approximation (GGA), which typically underbinds and overestimates lattice constants. Furthermore, our convergence study represents an idealized lattice at 0 K without zero-point vibrational energy, whereas experimental measurements naturally incorporate thermal expansion at room temperature.

2.2 β-cristobalite (SiO_2)

Our structural model employs the ideal β-cristobalite polymorph of SiO_2 in order to avoid dealing with an internal parameter related to the position of O atoms. This model is the idealized version of the high-temperature phase of α-cristobalite [4]. It consists of silicon atoms occupying the position they take in the diamond structure, while the oxygen atoms form bridges between them. Its cubic symmetry makes it a useful computational model for simulating the amorphous silica (a-SiO_2) usually found in semiconductor devices. The justification for this substitution lies in the nature of the electronic structure of the material : the electronic properties of SiO_2 are primarily governed by its short-range atomic interactions, specifically the local tetrahedral geometry, rather than long-range periodicity. Consequently, the electronic DOS of β-cristobalite closely mirrors that of a-SiO_2, allowing us to accurately capture essential properties while maintaining a computationally convenient unit cell.

2.2.1 Cutoff energy

Figure 5 shows that from $E_{cut} = 36$ Ha onwards, the difference in energy per atom after each iteration becomes very small (e.g, lower than 0.5 mHa). Going to 40 Ha adds only 15 mHa/atom improvement, with significantly higher computational cost. We can also notice that β-cristobalite requires a higher E_{cut} than pure silicon due to the stronger localization of oxygen electronic states, which needs a larger basis set for accurate wavefunction representation. We choose $E_{cut} = 36$ Ha as the converged value.

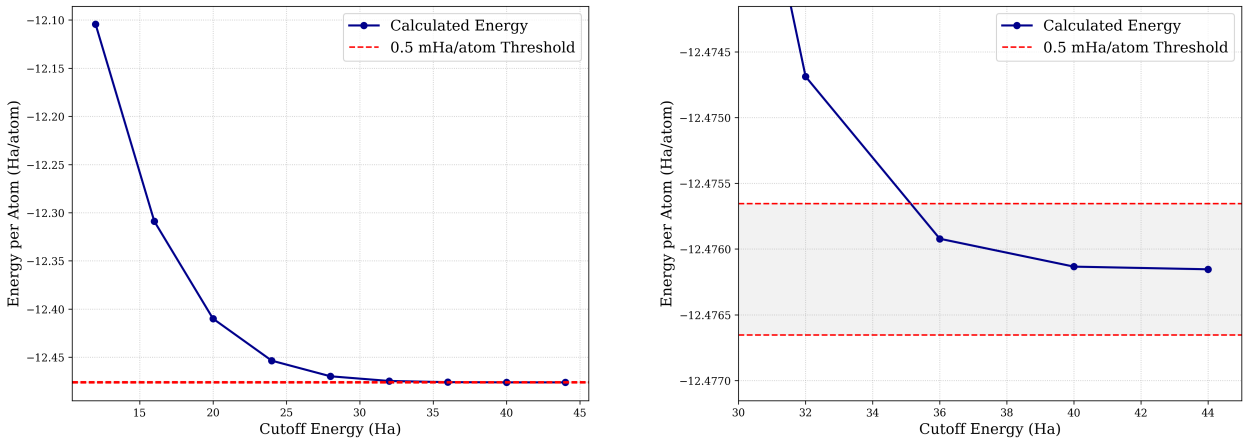


Figure 5: Convergence of the energy per atom as a function of the cut-off energy (E_{cut}) for β-cristobalite. The right plot shows a zoom-in on the region where the energy converges within the 0.5 mHa/atom threshold (shaded area).

2.2.2 K-point grid

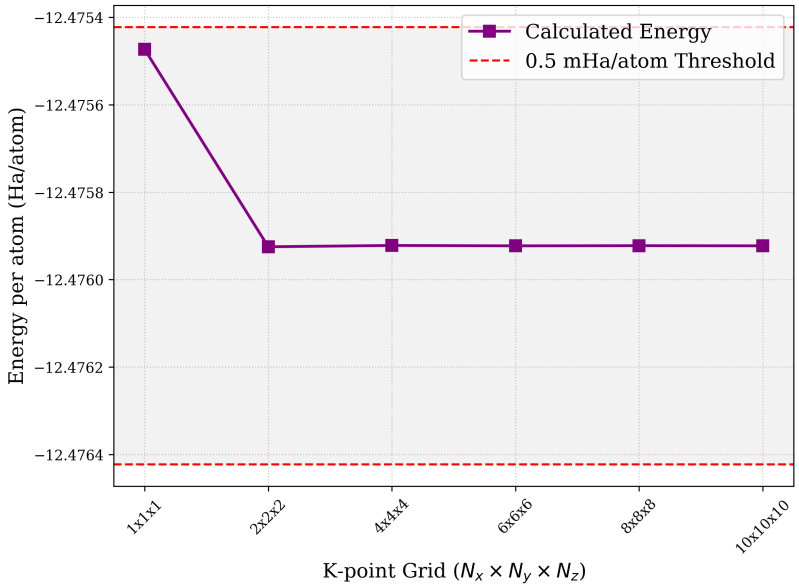


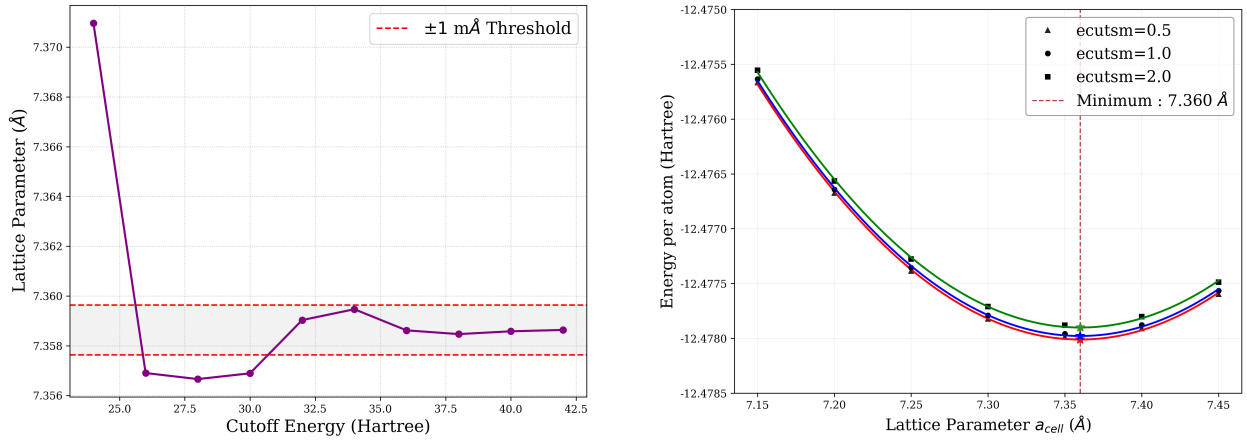
Figure 6: Convergence of the energy per atom as a function of the size of the k-point grid for β-cristobalite.

For the β-cristobalite bulk, we observed that a sampling of a single k-point ($1 \times 1 \times 1$) was sufficient for total energy to converge within the chosen threshold. Because the real-space unit cell of β-cristobalite is significantly larger than that of primitive silicon, its corresponding Brillouin

zone is proportionally smaller. Consequently, a coarser k-point mesh is sufficient to achieve an equivalent accuracy. Our convergence study, presented in figure 6, confirm that the total energy is fully converged at this grid size, allowing us to optimize computational efficiency without sacrificing accuracy.

2.2.3 Lattice parameter

The curve of figure 7a shows a different behavior from the silicon lattice parameter convergence study from figure 4. At low cutoff (24 Ha), the lattice parameter is clearly overestimated (about 7.370 Å), indicating the basis set is still incomplete. Increasing E_{cut} to 26 Ha causes a sharp drop to about 7.357 Å, which means the calculation is correcting that basis set error. Further correction is done from 28 Ha onwards such that the curve becomes nearly flat with only tiny oscillations after E_{cut}=36 Ha, for which the lattice parameter converges to 7.360 Å. Further cutoff energy increase only gives minimal benefit at a much higher computational cost.



(a) Using BFGS optimizer.

(b) Using the equation of state.

Figure 7: Convergence of the lattice parameter for β-cristobalite.

The lattice constant of figure 7a is obtained with the BFGS optimizer. To further validate the obtained value, a different approach was used, depicted in figure 7b. Here, the value of the lattice parameter is the variable and the value chosen as converged is the one for which the total energy per atom is the smallest. This approach is known as the equation of state (EOS). The three curves were generated using different choices of energy cutoff smearing, and although their overall shapes are similar, the position of the curve minimum shifts slightly around the red dashed line near 7.360 Å. Additionally, figure 7a and 7b clearly show that the resulting lattice parameter (7.360 Å) coincides for the two approaches and that our results are numerically stable, regardless of which smoothing value is picked.

This converged value can, once again, be compared to the experimental value (7.16 Å [5]) and the value reported by the Materials Project (7.4 Å [6]). The observed discrepancy can be attributed to the complex dynamic nature of the β-cristobalite phase. Unlike silicon, both our LDA calculation and the Materials Project value drastically overestimate the experimental lattice parameter. This occurs because β-cristobalite is a high-temperature phase characterized by massive dynamic disorder; in reality, the oxygen atoms constantly fluctuate around their average positions, bending the Si–O–Si bonds. However, when DFT simulates this phase as a static, 0 K ideal crystal with perfect cubic symmetry ($Fd\bar{3}m$), it forces the Si–O–Si bridging angles into a strictly linear 180° configuration. This artificial straightening of the bonds pushes the silicon atoms further apart and expands the calculated unit cell. Nevertheless, the theoretical trend between the two computational methods remains consistent with our silicon model: our LDA calculation yields a smaller lattice than the Materials Project's GGA-PBE functional due to the inherent overbinding tendency of LDA.

3 Electronic band structures and DOS

The parameters obtained in the convergence studies can now be applied to derive the band structures depicted hereafter, which confirm that E_{cut} and k-point grid are well-converged due to the smooth features and sharp peaks indicating proper parameter choices. The crystalline characteristics and converged values are synthetized in table 1.

Parameter	**Silicon (Si)**	**β-Cristobalite (SiO_2)**
Crystal system	Cubic	Cubic
Space group	$Fd\bar{3}m$ (227)	$Fd\bar{3}m$ (227)
Cutoff energy (`ecut`)	12.0 Ha	36.0 Ha
Brillouin zone sampling (`ngkpt`)	$4 \times 4 \times 4$	$1 \times 1 \times 1$
Converged lattice constant (a)	5.394 Å	7.36 Å

Table 1: Converged bulk parameters for Silicon and β-Cristobalite (SiO_2) obtained prior to interface construction.

The experimental band structure of silicon is shown in figure 8. The semiconducting behavior is pretty clear with an indirect band gap of 1.12 eV at room temperature. A quite large dispersion, translated as the curvature of the bands, can also be observed. This band structure can be used to criticize the results obtained herafter.

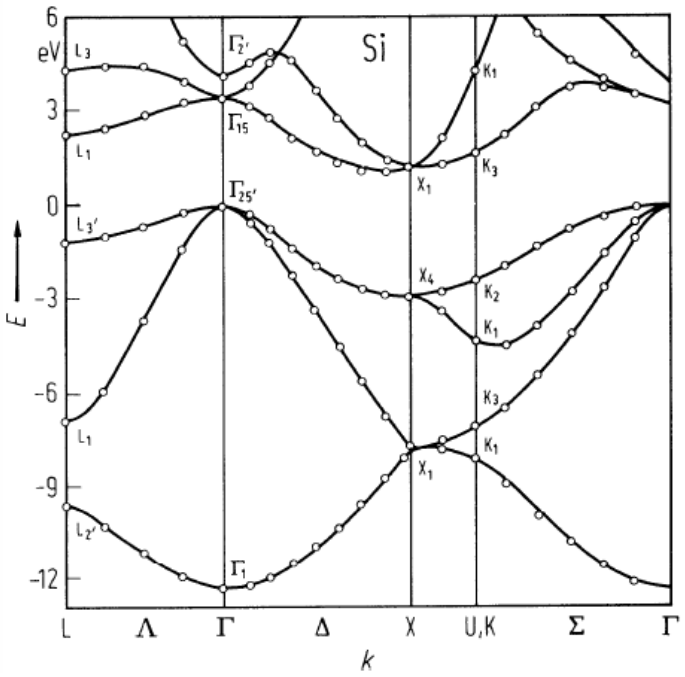


Figure 8: Silicon experimental band structure. The band gap is 1.12 eV at 300K. Adapted from [7].

In order to compute densities of states, occupation smearing is used to make Brillouin zone integrations numerically stable and smoother. On one hand, it can improve self-consistent field (SCF) convergence. Although both Si and SiO_2 are semiconducting and insulating materials with a finite band gap, SCF convergence can still be challenging because changes in the charge density alter the effective potential, which in turn shifts the Kohn–Sham eigenvalues and wavefunctions. These feedback effects may lead to oscillations or slow convergence, particularly when using large basis sets or poorly chosen mixing parameters. To mitigate these effects, smearing replaces sharp 0/1 occupations with smooth fractional occupations, which helps convergence. On the other hand, it

can reduce k-point noise. Because finite k-point grids can cause calculated quantities to look jagged, smearing is used to smooth the electron occupation around the Fermi level. This reduces noise in both the total energies and the density of states (DOS). For both DOS computations, Gaussian smearing is applied by setting `occopt` = 7.

Both band structures are plotted along high-symmetry directions in the Brillouin zone (W–L–Γ–X–W–K), while the DOS are displayed on the right as a function of energy. Figure 9 presents the key feature of the band structure of silicon, which is the presence of an indirect band gap separating the valence and conduction bands. The top of the valence band is located at the Γ point, whereas the bottom of the conduction band lies along the Γ–X direction rather than exactly at Γ. This confirms that silicon is an indirect band gap semiconductor.

It is important to note that the calculated value for the indirect band gap (0.45 eV) is significantly smaller than the experimental value (1.12 eV). This discrepancy is a well-known limitation of standard DFT methods, specifically the Local Density Approximation (LDA) used in this work, which tends to underestimate band gaps due to approximations in the exchange-correlation functional.

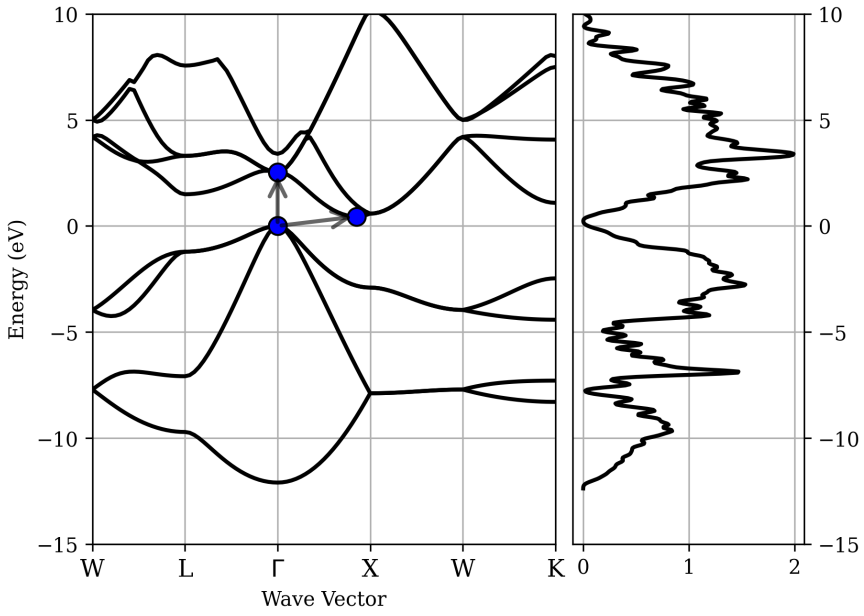


Figure 9: Computed band structure and DOS of silicon.

The DOS of silicon complements the band structure of figure 9 by showing the distribution of electronic states across the energy range. The DOS clearly vanishes in the band gap region, confirming the semiconducting nature of the material. Peaks in the DOS correspond to flatter bands in the band structure, where many states are concentrated in a narrow energy range.

The band diagram of β-cristobalite displayed in figure 10 also allows to confirm its wide-gap insulator nature, which explains its optical transparency to visible light (at room temperature, in quartz phase) and electrical insulating properties. However, the precise value of the band gap given by DFT-LDA (5.4 eV) is (as expected) underestimated compared to the experimental one (around 8.9 eV [8]). Also, the band diagram exhibits a typical oxide ceramic electronic structure, characterized by strongly localized states and weak band dispersion. The localization originates from the predominantly ionic nature of the Si–O bond: oxygen, being significantly more electronegative than silicon, attracts and retains the valence electrons. Consequently, the top of the valence band is mainly composed of O 2p orbitals that are spatially confined around the oxygen atoms, reflecting limited electron sharing compared to more covalent systems such as silicon.

The weak dispersion of the bands arises from the small orbital overlap between neighboring

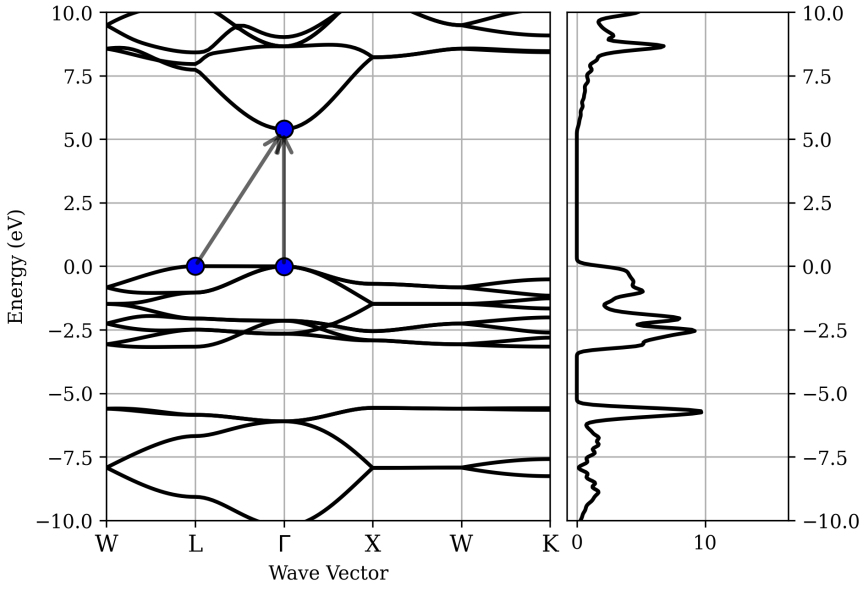


Figure 10: Computed band structure and DOS of β-cristobalite.

atoms. In β-cristobalite, the relatively large O–O distances and the absence of direct O–O bonding significantly reduce the overlap between O 2p orbitals. This leads to a low probability for electron hopping between sites, which in reciprocal space results in flat, weakly dispersive bands. In contrast, silicon exhibits stronger hybridization and greater electron delocalization, thus producing broader and more dispersive bands.

In conclusion, the DFT results correctly capture the qualitative electronic structure of silicon and β-cristobalite, including their indirect band gaps and overall band dispersion. However, the systematic quantitative underestimation of the band gap highlights the limitations of standard DFT-LDA for predicting excited-state properties of semiconducting and insulating materials. More advanced methods, such as *GW* corrections, are required for accurate band gap predictions.

4 Si-SiO_2 interface

Now that the bulk structures of Si and SiO_2 crystals are successfully built separately, our goal will be to join those crystals in a supercell by stacking SiO_2 on top of a Si slab. Ultimately, this will enable us to analyze the electronic properties at the Si/SiO_2 interface and, in particular, the macroscopic electrostatic potential allowing to extract the valence band offset (VBO) and conduction band offset (CBO).

4.1 Interface construction

The interface is modeled using a periodic slab approach, combining crystalline silicon with a silicon dioxide layer. A key challenge in building the interface lies in the significant lattice mismatch between the standard cubic cell of Si ($a = 5.394$ Å) and that of β-cristobalite ($a = 7.36$ Å). To resolve this without introducing destructive physical strain, the Si block is rotated by 45° in the xy-plane using a $\sqrt{2} \times \sqrt{2}R45°$ transformation matrix. This rotation changes the effective in-plane lattice parameter of the Si cell to its face diagonal ($5.394 \times \sqrt{2} \approx 7.68$ Å), which drastically reduces the mismatch. The SiO_2 slab is then strained in the xy-plane to perfectly match the rotated Si lattice vectors. Finally, the oxide layer is translated along the z-axis to establish an initial unrelaxed interfacial gap of 1.5 Å.

The resulting supercell, shown in figure 11, consists of three regions: a bulk-like silicon slab, an interfacial gap region and a SiO_2 layer. The silicon region contains 7 atomic layers to ensure bulk-like behavior away from the interface, while the oxide region contains 5 layers. 25 Å of vacuum have also been introduced along the z-axis in order to avoid spurious interactions between neighboring cells.

Moreover, it is important to implement proper chemical termination at the boundaries. Indeed, dangling bonds can introduce mid-gap states that can be detrimental to the determination of band edges. Therefore, these dangling bonds are mitigated through hydrogen passivation by introducing hydrogen atoms (white atoms in figure 11) at the boundaries of our supercell. The resulting system consists of 68 atoms (36 Si, 20 O, 12 H).

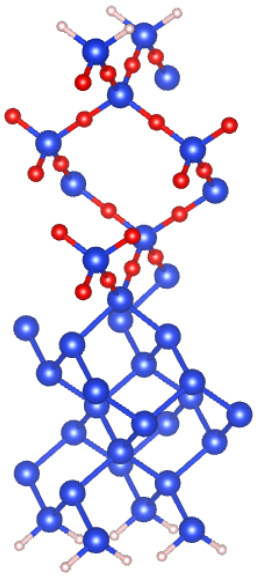


Figure 11: VESTA visualization of the initial, unrelaxed supercell.

4.2 Structural relaxation

Once the initial supercell is built, structural relaxation can be performed. The objective of this step is to minimize atomic forces while preserving the overall macroscopic geometry of the interface. The atomic relaxation is carried out using the BFGS algorithm, with a convergence criterion of 5.0×10^{-5} Hartree/Bohr (2.6 meV/Å) for the maximum atomic force.

The silicon substrate is much thicker and mechanically dominates the system. The oxide layer forms while remaining bonded to the Si surface. However, at the interface, atoms must share a common periodicity in the plane of the interface. As a result, the interfacial SiO_2 cannot adopt its fully relaxed bulk lattice parameters. Instead, it is strained to match the in-plane dimensions dictated by the silicon substrate. In order to maintain this epitaxial strain and avoid artificial structural redistributions, the simulation cell dimensions are strictly locked, thus fixing the in-plane lattice parameters and enforcing a common periodicity across the interface. This allows the calculations to capture the structural and electronic effects arising from lattice mismatch. Moreover, Brillouin zone sampling is adapted to the 2D slab geometry using a $2 \times 2 \times 1$ Monkhorst-Pack k-point grid with a plane-wave cutoff energy of 36 Ha. Relaxation results are shown in figure 12.

When the initial supercell was built, the crystals of Si and SiO_2 were initially placed in positions that have not yet accounted for their new neighbors. Because the lattice constants of Si and SiO_2 do not match perfectly, two issues appeared : firstly, some atoms at the Si/SiO_2 boundary might have unpaired electrons, and secondly, the initial bonds across the interface were often unnaturally long leading to weak bonds, which significantly alters the electronic energy levels.

At the unrelaxed interface, those dangling bonds and stretched atomic distances created new energy levels within the band gaps of Si and SiO_2. Those defects states make the system behave like a metal in the simulation, which is an issue because a tiny change in the potential during one step of the SCF iterations can cause a massive amount of electronic charge to suddenly jump from one side of the interface to the other. This phenomenon is known as artificial charge sloshing. In the next step of the iteration, the computer tries to correct this and the charge "sloshes" back. As a result, the simulation never settles down and fails to converge.

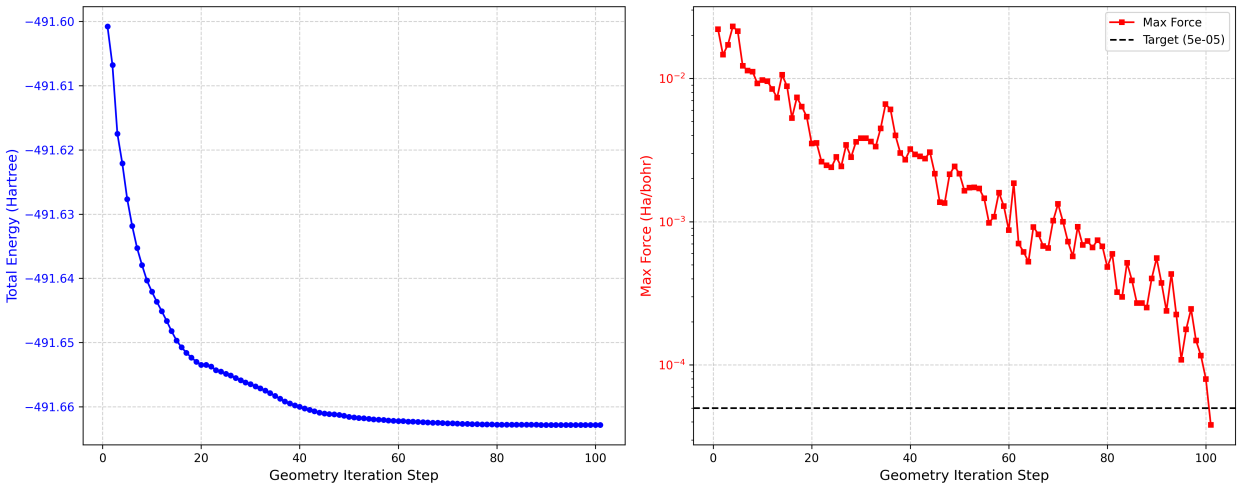


Figure 12: Structural relaxation profile of the simulated Si/SiO2 interface. The left panel demonstrates the smooth minimization of the system's total energy. The right panel displays the evolution of the maximum atomic force on a logarithmic scale. The relaxation successfully terminates when the maximum force drops below the strict convergence criterion of 5e-05 Ha/Bohr (dashed black line), reached after 101 iteration steps.

To guarantee stable SCF convergence during the geometry optimization, a Fermi-Dirac smearing with a width of 0.01 Ha is applied to the occupation numbers that allows states to be fractionally occupied. Indeed, now the probability f_i of a state being occupied is not 0 or 1 but given by:

$$f_i = \frac{1}{\exp\left(\frac{\epsilon_i - \mu}{k_B T}\right) + 1} \tag{3}$$

with μ the chemical potential, ϵ_i the energy of the state i and $k_B T$ the smearing width. This technique makes the energy surface smoother instead of forcing an electron to be in a specific state. As a result, it prevents the SCF from overreacting to tiny potential changes and atoms can now move into their proper relaxed positions without any issues. Once the relaxation was completed, the "metallic" defect states disappeared and the system returned to being a semiconductor/insulator interface. This use of smearing enabled the successful relaxation of the interface, whose final structure is compared to the initial structure in figure 13.

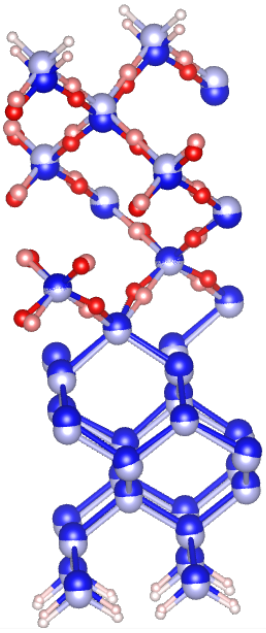


Figure 13: Comparative atomic visualization of the Si/SiO_2 interface structure, before and after relaxation. The final relaxed geometry (vibrant dark blue Si, bright red O) is superimposed over the initial, unrelaxed coordinates (faded blue Si, light pink O). Significant atomic rearrangements appears in the SiO_2 network to relieve internal network strain.

Another issue can be put forward in the vacuum region related to a specific type of charge sloshing that occurs over long distances. Indeed, the density of charge in the vacuum region should be zero, but because of the periodic nature of the calculation, tiny charges can appear in the vacuum.

These charges create long-range electric fields that can cause the electron density to oscillate wildly back and forth across the vacuum gap during each iteration. To solve this, a modified dielectric function (`diemac` = 10.0, `diemix` = 0.5) was implemented to dampen long-range charge oscillations across the vacuum so they don't spiral out of control. After relaxation, we verify in table 2 that the optimized structure exhibits stable Si–O bond lengths on the order of 1.6 Å and does not contain nonphysical distortions.

Table 2: Calculated bond lengths and bond angles for selected atoms pairs and triplets at the Si/SiO_2 interface and within the deep bulk regions. Corresponding atom numbers are indicated in figure 14.

Bond Lengths (Å)		**Bond Angles (deg)**	
Bond	Distance	Angle	Value
Si_{28}–O_8	1.635	O_8–Si_{28}–O_5	121.2
Si_{27}–O_5	1.646	O_3–Si_{26}–O_2	116.7
Si_{26}–O_4	1.635	O_{20}–Si_{36}–$O_{19_{bulk}}$	114.6
Si_{25}–O_1	1.646	O_5–Si_{27}–O_4	111.6
Si_{36}–O_{20}(bulk)	1.620		
Si_{16}–Si_{27}	2.385	Si_{35}–O_{18}–$Si_{34_{bulk}}$	170.7
Si_{14}–Si_{27}	2.385	Si_{27}–O_4–Si_{26}	154.9
Si_{16}–Si_{21}	2.352		
Si_{17}–$Si_{4_{bulk}}$	2.332	Si_5–Si_4–$Si_{3_{bulk}}$	110.6

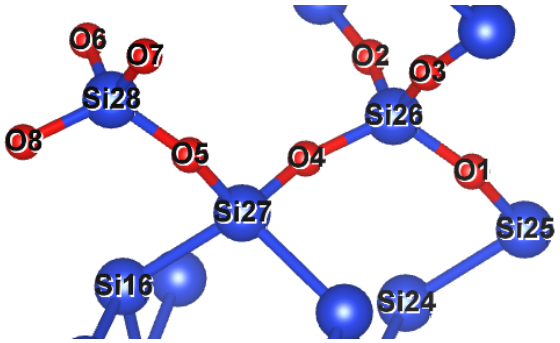


Figure 14: Close-up view of the relaxed interface with labeled atoms.

4.3 Electrostatic potential

Now that our interface is well-relaxed, thus avoiding false contributions to electronic properties across the interface, we can proceed to the calculation of the total electrostatic potential. To extract meaningful macroscopic properties, the 3D total local potential (ionic, Hartree and exchange-correlation components) is averaged over the xy-plane at each point z to yield a 1D planar average. Looking at the grey curve in figure 15, the rapid oscillations corresponds to the atomic structure. More precisely, the deep wells are the locations of atomic layers where the positive charge creates a strong local potential.

However, the absolute energy scale is ill-defined because the planar average electrostatic potential depends on the specific periodic arrangement of charges in the crystals. To overcome this, the macroscopic average technique is employed by applying a moving average with a window size equal to the bulk silicon unit cell parameter, namely 5.394 Å. As a result, the atomic-scale oscillations

caused by atomic cores and bonding are smoothed out. This procedure allows to systematically filter out the highly localized atomic oscillations and reveals the ”bulk-like” potential of each material.

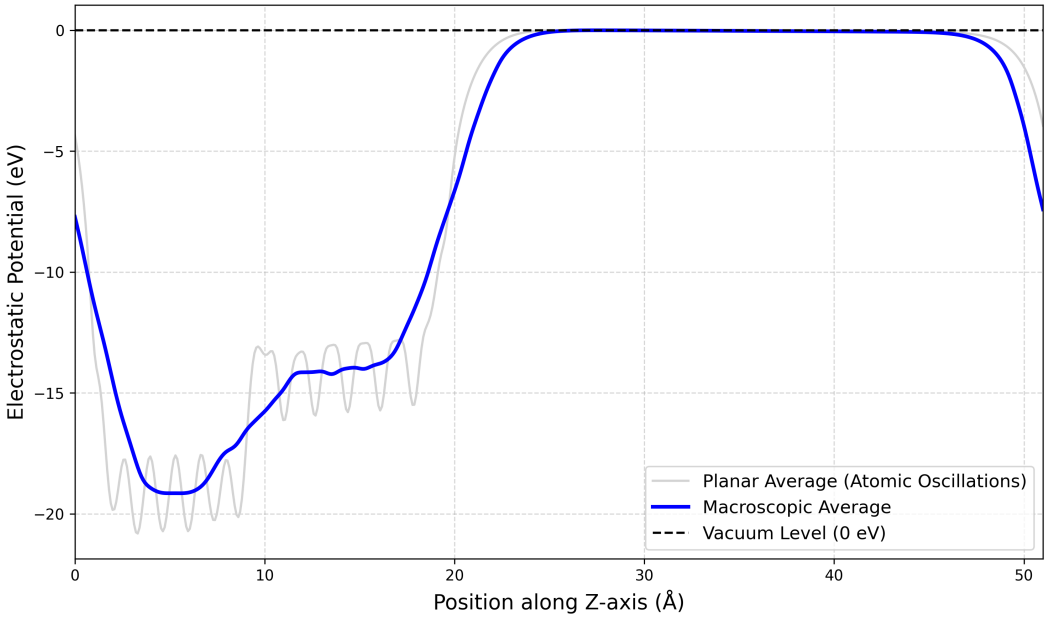


Figure 15: Planar (grey) and macroscopic (blue) averages of the total electrostatic potential along the z-axis of the Si/SiO_2 supercell. The macroscopic average reveals distinct thermodynamic plateaus for the bulk Si, bulk SiO_2, and the vacuum region (which is shifted to 0 eV).

Based on the macroscopic average potential, the supercell can be divided into 3 distinct regions. The bulk silicon region, located around 0 to 8 Å, is characterized by regular, periodic oscillations in the grey line and a macroscopic potential plateau at approximately -19 eV. Around 9-10 Å , the interface manifests as a transition region where the potential rises sharply because of the chemical bonding between Si and O atoms. The SiO_2 region between 11 and 17 Å exhibit a shallower plateau at -14 eV. Furthermore, the macroscopic potential features a flat plateau in the vacuum region around 20 to 50 Å, demonstrating that the supercell is well-isolated and free of spurious, artificial macroscopic electric fields across the boundaries. Note that, without the modified dielectric function, the macroscopic potential in this region might have wiggled or failed to reach a steady slope because of tiny charge long-range oscillations.

Physically, having a plateau in the macroscopic average essentially means that we have moved in a region far enough from the interface to reach a state of electrostatic equilibrium, where the local environment of an atom is identical to what it would be inside an infinite crystal of that same material. We call such regions ”bulk-like”. Therefore, the presence of those plateaus confirms that the respective material layers are sufficiently thick to exhibit bulk-like behavior away from the boundary, allowing for accurate calculations of the energy barriers that govern modern electronic devices behavior. The difference between the two bulk plateaus corresponds to the macroscopic average potential shift across the interface, introducing a potential step of approximately $\Delta V = V_{SiO_2} - V_{Si} = 5.0$ eV. This value is a critical input for calculating the VBO and CBO, which are used to determine how electrons and holes move across the Si/SiO_2 junction.

It is important to note the referencing convention used for the electrostatic potential in figure 15. In periodic DFT calculations, the absolute value of the electrostatic potential ($V_{\text{ions}} + V_{\text{Hartree}} + V_{XC}$) bears no physical meaning, as it is defined only up to an arbitrary integration constant. Only potential differences (ΔV) across the system are physically relevant. Therefore, we have aligned the flat macroscopic plateau of the isolated vacuum region to 0 eV to serve as a consistent baseline.

4.4 Band offsets

The determination of the band offsets at the Si/SiO_2 interface requires coupling the electronic band profiles of the bulk systems to the macroscopic electrostatic potential profile extracted from the supercell. Because we are modeling periodic systems, the calculated electron eigenvalues are unique only up to an arbitrary constant. Different simulation setups or pseudopotentials can yield varying absolute Valence Band Maximum (VBM) and Conduction Band Minimum (CBM) values simply due to their internal zero-energy conventions. Consequently, these raw eigenvalues are not physically well-defined on an absolute scale, though their differences (the band gaps) are well-defined.

To calculate the VBO (ΔE_v), we must bridge the isolated bulk calculations using the interfacial potential step. Indeed, the calculated VBM values for bulk Si (4.88 eV) and bulk SiO_2 (-1.55 eV) are, by convention, aligned to the average electrostatic potential ($V_{\text{ions}} + V_{\text{Hartree}} + V_{XC}$) of their respective unit cells, specifically the $G = 0$ component of the potential. By extracting the macroscopic potential step across the interface from the supercell calculation ($\Delta V_{\text{Si/SiO}_2} = -5.00$ eV), we can align these local references to a common baseline. Therefore, the VBO can be expressed as:

$$\Delta E_v = E_{\text{VBM,Si}} - E_{\text{VBM,SiO2}} + \Delta V_{\text{Si/SiO}_2} = 4.88 - (-1.55) + (-5.00) = 1.43 \text{ eV} \tag{4}$$

Once the VBO is established, the CBO (ΔE_c) can be derived from the calculated bandgaps (E_g) of the two materials:

$$\Delta E_c = E_{g,\text{SiO}_2} - E_{g,\text{Si}} - \Delta E_v = 5.39 - 0.45 - 1.43 = 3.51 \text{ eV} \tag{5}$$

A comparison of our results for band offsets with similar studies and experimental results is shown in table 3.

Table 3: Comparison of calculated and experimental VBO and CBO for the Si/SiO_2 interface.

Source / Methodology	**VBO (eV)**	**CBO (eV)**
Our work (LDA)	1.43	3.51
LDA [9]	2.2	3.3
LDA-1/2 [9]	4.00	3.00
Experimental [8]	4.50	3.30

The observed discrepancy between our calculated results and the experimental values in Table 3 stems from the well-known limitations of standard Kohn-Sham DFT. Because standard DFT is inherently formulated for ground-state electronic densities, it routinely underestimates excited-state properties, including band gaps. Specifically, the semilocal functionals employed in this work suffer from self-interaction error (SIE), where an electron unphysically interacts with its own generated electrostatic field. This artificial repulsion destabilizes the occupied valence states, shifting the valence band maximum (VBM) to higher energies. Consequently, the absolute positions of the VBM are distorted, compromising the accuracy of the subsequent VBO and CBO calculations.

To improve the agreement with experiment, we see in table 3 that the method LDA-1/2 has much better values for the band offsets. It does so by modifying the atomic pseudopotentials to mimic self-energy corrections by removing half an electron from the valence shell, providing GW-like accuracy at a fraction of the computational cost.

5 Conclusion

This report presented a first-principles study of the Si/SiO2 interface, a fundamental component governing charge transport and quantum tunneling in modern MOSFET devices. Using DFT-LDA, we systematically modeled the bulk properties of both silicon and β-cristobalite, followed by the

structural and electronic characterization of their interface. Initial convergence studies ensured robust computational parameters, yielding bulk lattice constants and electronic band structures consistent with the theoretical expectations of the LDA functional. The characteristic underestimation of band gaps—yielding 0.45 eV for Si and 5.39 eV for SiO2—and the deviations in lattice parameters were observed and contextualized, particularly regarding the dynamic, high-temperature nature of the β-cristobalite phase. The construction of the interface supercell required geometric alignment via a 45-degree in-plane rotation of the silicon slab to minimize lattice mismatch. Structural relaxation was performed to relieve internal strain and eliminate artificial defect states. The computation of the macroscopic electrostatic potential yielded a potential step of 5 eV at the interface, allowing for the computation of the VBO (1.43 eV) and CBO (3.51 eV). A comparison with similar works and experimental results allowed to highlight the defects of our work and the limitations of DFT. Future work might use *GW* corrections to predict band gaps more accurately, as well as more advanced/recent exchange-correlations functionals for a more precise calculation of the VBM.

6 Personal contributions

- **DFT calculations:** Calculations were performed by both students to cross-check the obtained values.
- **Results processing and plotting using abipy:** Plots were mostly done by Jean-Lou, as well as VESTA visualizations with the exception of figure 7b made by Lucas.
- **Report writing:** The writing was a collaborative work cross-checked several times by both students.
- **Literature gathering:** Literature was gathered by both students.

Acknowledgments

We would like to sincerely thank our teaching assistant Wei Chen for his valuable and trustworthy advice and help during the project. We would also like to thank professors Gian-Marco Rignanese and Samuel Poncé for their important feedback which allowed us to improve the quality of this report. Finally, we would like to acknowledge that AI tools ChatGPT and Gemini were used during this project for several purposes : Python code generation for raw data analysis and plotting, help in understanding physical and numerical concepts, rephrasing or clarification of text.

A Appendices

A.1 Input files

A.1.1 Convergence study of silicon cutoff energy

```
# Silicon (Si) convergence study on ecut
# Si in the diamond cubic structure

ndtset 10

# We fix the lattice constant (use the expt one) and vary ecut only
acell 3*5.431 angstrom

ecut: 8.0
ecut+ 2.0

#Definition of the atom types
```

```
ntypat 1          # There is only one type of atom
znucl 14

pp_dirpath "$ABI_PSPDIR"        # path to the directory were
                                # pseudopotentials for tests are stored

pseudos "Psdj_nc_sr_04_pw_std_psp8/Si.psp8"
                                # Name and location of the pseudopotential


# Definition of the primitive FCC diamond structure (Much faster!)
rprim  0.0  0.5  0.5
       0.5  0.0  0.5
       0.5  0.5  0.0

natom 2                 # Only 2 atoms needed for primitive cell
typat 1 1               # Both are Silicon
xred                    # Use fractions so they scale with acell!
   0.00  0.00  0.00     # Atom 1 at the origin
   0.25  0.25  0.25     # Atom 2 at 1/4 diagonal


#Definition of the k-point grid

kptopt 1          # Option for the automatic generation of k points, taking
                  # into account the symmetry
ngkpt 4 4 4       # Denser k-point grid for accurate total energies
nshiftk 4         # of the reciprocal space (that form a BCC lattice !),
                  # repeated four times, with different shifts :
shiftk 0.5 0.5 0.5
       0.5 0.0 0.0
       0.0 0.5 0.0
       0.0 0.0 0.5

#Definition of the SCF procedure
nstep 50          # Maximal number of SCF cycles
toldfe 1.0d-10    # Force the electrons to converge to a tighter tolerance than the atoms
nband 8           # Number of bands
diemac 12.0       # Dielectric preconditioner
```

A.1.2 Convergence study of β-cristobalite lattice parameter

```
    # Beta-Cristobalite (SiO2) optimized acell convergence by varying ecut
# This test finds the equilibrium lattice parameter
ndtset 10

# We fix the lattice constant (use the expt one) and vary the ecut only
acell 3*7.16 angstrom

# 2. Ecut Sweep: Start lower and move past 36.0 Ha (converged value from 1_ecut)
# This ensures we see the "flattening" of the lattice parameter
ecut: 24.0
ecut+ 2.0
```

```
#Definition of the atom types
ntypat 2          # Si and O
znucl 14 8

pp_dirpath "$ABI_PSPDIR"        # This is the path to the directory were
                                # pseudopotentials for tests are stored

pseudos "Psdj_nc_sr_04_pw_std_psp8/Si.psp8, Psdj_nc_sr_04_pw_std_psp8/O.psp8"

# Definition of the primitive FCC structure
rprim  0.0  0.5  0.5
       0.5  0.0  0.5
       0.5  0.5  0.0

#Definition of the atoms
natom 6                # 6 atoms needed for primitive cell
typat 1 1 2 2 2 2      # two Si + four O
xred
  0.000  0.000  0.000
  0.250  0.250  0.250
  0.125  0.125  0.125
  0.625  0.125  0.125
  0.125  0.625  0.125
  0.125  0.125  0.625

# Definition of the k-point grid
kptopt 1
ngkpt 4 4 4       # Sufficient according to 2_kpt
nshiftk 4
shiftk 0.5 0.5 0.5
       0.5 0.0 0.0
       0.0 0.5 0.0
       0.0 0.0 0.5

# Definition of the SCF procedure
nstep 50
toldfe 1.0d-10
nband 24          # Must be 24+ to handle the 32 valence electrons
diemac 6.0        # Insulator setting for SiO2

# --- AUTOMATIC CELL OPTIMIZATION ---
optcell  2         # Optimize cell volume
ionmov   2         # Move atoms using BFGS algorithm
ntime    50        # Max steps for the box to "find" its shape

# Plane-Wave Smearing
ecutsm   0.5
dilatmx  1.10      # Give Oxygen a bit more room (10%) to expand if needed
tolmxf   2.0d-5
```

A.1.3 Band structure and DOS of β-cristobalite

```
    # Beta-Cristobalite Band Structure and Density of States
ndtset 3

# --- COMMON PARAMETERS (Geometry & Convergence) ---
# Using converged lattice parameter
acell 3*7.36 angstrom
rprim  0.0  0.5  0.5
       0.5  0.0  0.5
       0.5  0.5  0.0

ntypat 2
znucl 14 8
natom 6
typat 1 1 2 2 2 2
xred
  0.000  0.000  0.000
  0.250  0.250  0.250
  0.125  0.125  0.125
  0.625  0.125  0.125
  0.125  0.625  0.125
  0.125  0.125  0.625

pp_dirpath "$ABI_PSPDIR"
pseudos "Psdj_nc_sr_04_pw_std_psp8/Si.psp8, Psdj_nc_sr_04_pw_std_psp8/O.psp8"

ecut 36.0                 # Converged plane-wave cutoff
nband 24                  # 16 filled bands + 8 empty conduction bands

# ============================================================================
# DATASET 1: SCF Ground State (Calculate the Density)
# ============================================================================
kptopt1 1
ngkpt1 4 4 4              # Converged grid from previous k-point study
nshiftk1 4
shiftk1 0.5 0.5 0.5
        0.5 0.0 0.0
        0.0 0.5 0.0
        0.0 0.0 0.5

toldfe1 1.0d-10
prtden1 1                 # Save the electron density
diemac1 6.0               # Preconditioner for wide-gap insulators!

# ============================================================================
# DATASET 2: NSCF Band Structure
# ============================================================================
iscf2 -2
getden2 -1
tolwfr2 1.0d-12

kptopt2 -5                # 5-segment FCC path
```

```
ndivk2 5*20
kptbounds2
  0.500  0.250  0.750     # W
  0.500  0.500  0.500     # L
  0.000  0.000  0.000     # Gamma
  0.500  0.000  0.500     # X
  0.500  0.250  0.750     # W
  0.375  0.375  0.750     # K

# ============================================================================
# DATASET 3: NSCF Density of States (DOS)
# ============================================================================
iscf3 -2
getden3 -2
tolwfr3 1.0d-12

kptopt3 1
ngkpt3 8 8 8              # Dense uniform grid for a smooth curve
nshiftk3 4
shiftk3 0.5 0.5 0.5
        0.5 0.0 0.0
        0.0 0.5 0.0
        0.0 0.0 0.5

prtdos3 1
occopt3 7                 # Gaussian smearing
tsmear3 0.01
```

A.1.4 Relaxation of the interface

```
ndtset 3

autoparal 1
npfft 1

ionmov1  2        # Let the interface atoms move to find equilibrium
optcell1 0        # Lock the box dimensions
ntime1   20      # Maximum number of relaxation steps
tolmxf1  5.0d-5   # Force tolerance to stop the outer loop (atoms)
toldfe1  1.0d-6   # Energy tolerance to stop the inner loop (electrons)

# THIS PART WAS NOT USED FOR THE POTENTIAL CALCULATION, ANOTHER RUN WAS NEEDED
prtvha2  1        # Print the 3D Hartree Potential map
tolvrs2  1.0d-10  # Strict energy convergence for the final potential

prtdos3  3        # Project the DOS onto every individual atom
tolwfr3  1.0d-10  # Wavefunction tolerance for DOS calculation
#END OF THE UNUSED PART

ecut     36.0     # Converged cutoff energy
nstep    100    # Max SCF steps per cycle
natom  68
nsppol 1
```

```
diemac 10.0
diemix 0.5
iscf  7
occopt 3      # Fermi-Dirac smearing for transient metallic states
tsmear 0.01   # Smearing temperature in Hartree

# k-point grid
kptopt   1
ngkpt    2 2 1
nshiftk  1
shiftk   0.0 0.0 0.0

#Definition of the unit cell
acell
1.00000000000000 1.00000000000000 1.00000000000000 Angstrom

rprim
5.39400000000000 -5.39400000000000 0.00000000000000
5.39400000000000 5.39400000000000 0.00000000000000
0.00000000000000 0.00000000000000 32.00000000000000

chkprim 0 # Allow non-primitive cells

#Definition of the atom types
ntypat 3
znucl 8 14 1
pp_dirpath "$ABI_PSPDIR"
pseudos "Psdj_nc_sr_04_pw_std_psp8/O.psp8, Psdj_nc_sr_04_pw_std_psp8/Si.psp8,
        Psdj_nc_sr_04_pw_std_psp8/H.psp8"

#Enumerate different atomic species
typat
 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
 2 2 2 2 2 2 1 1 1 1 2 2 1 1 1 1 2 2 1 1
 1 1 2 2 1 1 1 1 2 2 1 1 2 2 1 1 3 3 3 3
 3 3 3 3 3 3 3 3

#Definition of the atoms
xcart
 ...

 chkexit 1
```

A.2 Pseudopotential references

Both bulk and interface calculations in this work utilized the following pseudopotentials:

- **Element: Si**, URL: `http://www.pseudo-dojo.org/`, Atomic config: $3s^2 3p^2$, Type: NC, XC functional: GGA-PBE
- **Element: O**, URL: `http://www.pseudo-dojo.org/`, Atomic config: $2s^2 2p^4$, Type: NC, XC functional: GGA-PBE
- **Element: H**, URL: `http://www.pseudo-dojo.org/`, Atomic config: $1s^1$, Type: NC, XC functional: GGA-PBE

Note on Functional Consistency: While the header metadata of the PseudoDojo .psp8 files indicates they were generated using the GGA-PBE functional (pspxc = 11), our bulk and interfacial ABINIT calculations were explicitly forced to execute using the LDA Perdew-Wang 92 functional (ixc = -1012). This slight functional inconsistency between the core generation and the valence interaction was noted, but the converged silicon lattice parameter of 5.394 Å confirms that the system behaved in accordance with expected LDA results.

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